22q11.2 Deletion Syndrome: A Comprehensive Disease Characteristics Report

Summary

22q11.2 Deletion Syndrome (22q11.2DS) — historically fragmented into DiGeorge syndrome, velocardiofacial (Shprintzen) syndrome, and conotruncal anomaly face syndrome — is the **most common human chromosomal microdeletion syndrome**, affecting approximately **1 in 4,000 live births** (PMID: 35645294). It is caused by a recurrent hemizygous deletion of chromosome 22q11.2, most commonly a **~3 Mb "typically-deleted region" (TDR)** spanning low-copy-repeat blocks LCR22A–D and encompassing **~106 genes** (PMID: 32117416). Haploinsufficiency of the T-box transcription factor **TBX1** is considered the single greatest contributor to the physical phenotype, disrupting development of the pharyngeal apparatus and neural crest (PMID: 23799583).

The syndrome is a **lifelong, incurable, multisystem disorder** with extraordinarily variable expressivity. Its cardinal features derive from a developmental failure of the pharyngeal arches and neural crest: **conotruncal congenital heart disease** (40–50% of patients, and the leading cause of pediatric mortality), **thymic hypoplasia with T-cell immunodeficiency**, **parathyroid hypoplasia with hypocalcemia**, **palatal/velopharyngeal anomalies**, and **characteristic craniofacial dysmorphism**. Superimposed on these is a heavy **neuropsychiatric burden**: intellectual disability, a **20–30% lifetime risk of schizophrenia**, autism-spectrum traits, ADHD, and a markedly elevated risk of **early-onset Parkinson's disease** (PMID: 42585902; PMID: 39918054). Genes on the intact homolog (notably **COMT** Val158Met) and rare damaging variants in **chromatin-regulatory modifier genes** shape the severity of both cardiac and neurocognitive outcomes.

Inheritance is **autosomal dominant**, but ~90% of deletions arise **de novo** through non-allelic homologous recombination between the flanking low-copy repeats; an affected individual has a ~50% chance of transmitting the deletion (PMID: 14708107). Diagnosis relies on **chromosomal microarray, FISH, or MLPA**, with prenatal cell-free DNA screening now achievable at high positive predictive value when fetal-fraction amplification is used (PMID:

38622914). Management is **organ-directed and guideline-based** (cardiac surgery, calcium/vitamin D supplementation, immune surveillance and thymus transplantation for the rare complete-athymia subset, and lifelong psychiatric monitoring).

Disease Overview and Identifiers

Section 1 — Disease Information. 22q11.2DS is a contiguous gene deletion syndrome affecting multiple organ systems derived from the embryonic pharyngeal apparatus. It unifies several previously separate clinical entities. The information in this report is drawn from **aggregated, disease-level resources** (OMIM, Orphanet, consensus clinical practice guidelines, and cohort studies) rather than from individual EHR records.

Resource	Identifier
MONDO	MONDO:0008644 (22q11.2 deletion syndrome)
OMIM	#188400 (DiGeorge syndrome); #192430 (velocardiofacial syndrome)
Orphanet	ORPHA:567
ICD-10	D82.1 (DiGeorge syndrome)
ICD-11	LD44.5 / 4A00.1Y (DiGeorge anomaly)
MeSH	D004062 (DiGeorge Syndrome) / 22q11 Deletion Syndrome

Common synonyms: DiGeorge syndrome (DGS), velocardiofacial syndrome (VCFS), Shprintzen syndrome, conotruncal anomaly face syndrome (CTAF/Takao syndrome), CATCH-22 (a now-deprecated mnemonic: Cardiac defects, Abnormal facies, Thymic hypoplasia, Cleft palate, Hypocalcemia), and thymic hypoplasia.

Key Findings

Finding 1 — The 22q11.2 microdeletion and TBX1 haploinsufficiency are the molecular cause

The syndrome is caused by a **hemizygous 22q11.2 microdeletion**. Approximately **85–90% of patients carry the ~3 Mb "typically-deleted region" (TDR)** bounded by low-copy repeats **LCR22A–LCR22D**, producing haploinsufficiency of **~106 genes**, including coding RNAs, non-coding RNAs, and pseudogenes (PMID: 32117416). A smaller ~1.5 Mb proximal deletion (LCR22A–B) accounts for most craniofacial phenotypes (PMID: 35645294).

Among the deleted genes, **TBX1** — a T-box transcription factor — is the **greatest single contributor** to the disorder. As the review by Scambler and colleagues states: *"Although up to 50 genes may be deleted, it is haploinsufficiency of the transcription factor TBX1 that is thought to make the greatest contribution to the disorder. Mouse embryos are exquisitely sensitive to varying levels of Tbx1 mRNA, and Tbx1 is required in all three germ layers of the embryonic pharyngeal region for normal development"* (PMID: 23799583). The dose-sensitivity of the developing embryo to TBX1 is the molecular root of the phenotype.

- **Ontology:** Gene HGNC:11592 (TBX1); MONDO:0008644.

Finding 2 — Neural crest / pharyngeal apparatus disruption produces the cardinal malformations

TBX1 acts in the **pharyngeal ectoderm, endoderm, and mesoderm**, and also directs **non-cell-autonomous signaling between the pharyngeal surface ectoderm and rostral neural crest**. It interacts with the **fibroblast growth factor (FGF)**, **retinoic acid**, **CTNNB1/β-catenin (Wnt)**, and **bone morphogenetic protein (BMP)** pathways (PMID: 23799583). Disruption of neural crest development leads directly to the cardinal congenital malformations: *"Neural crest development is usually impacted leading to various cardiovascular defects including Tetralogy of Fallot and truncus arteriosus. It can also cause palatal defects, especially velopharyngeal deficiency"* (PMID: 40038168).

- **Ontology:** GO:0014032 (neural crest cell development); GO:0060021 (roof of mouth development); UBERON:0002539 (pharyngeal arch); CL:0000333 (migratory cranial neural crest cell).

Finding 3 — 22q11.2DS confers a 20–30% lifetime schizophrenia risk and broad neuropsychiatric burden

22q11.2DS is one of the strongest known genetic risk factors for psychiatric illness: *"22q11.2 deletion syndrome (22q11DS) confers a 20-30% lifetime risk of schizophrenia, with negative symptoms associated with poorer outcomes and higher psychosis risk"* (PMID: 42585902). Patients show elevated positive/disorganized symptoms, negative-motivational and negative-expressive dimensions, autism-spectrum traits, ADHD, and global neurocognitive impairment. Widespread brain structural alterations are documented across large consortia (ENIGMA-22q).

At the cellular level, genes within the deletion show a **strong expression bias toward VIP-expressing, caudal-ganglionic-eminence (CGE)-derived GABAergic interneurons**: *"genes within 22q11.2 deletions, known to confer a high risk for SCZ, ASD, and cognitive impairment, showed a strong expression bias toward vasoactive intestinal peptide-expressing cells (VIP+) among CGE subtypes"* (PMID: 40236134), implicating a specific vulnerable interneuron population.

- **Ontology:** HP:0100753 (schizophrenia); HP:0000717 (autism); HP:0007018 (ADHD); CL:0000617 (GABAergic interneuron); CL:4023070 (VIP GABAergic cortical interneuron).

Finding 4 — The deletion is common (~1 in 4,000 births) and frequently under-recognized

The 22q11.2 deletion *"is one of the most common genetic microdeletions, affecting approximately 1 in 4000 live births in humans"* (PMID: 35645294). It is a leading genetic cause of conotruncal congenital heart disease: in a cohort of adults with congenital heart disease, *"In Tetralogy of Fallot, 4.3% had 22q11.2 deletion, mostly diagnosed in childhood; among those without prior diagnosis, most were neither clinically suspected nor tested in adulthood, suggesting under-recognition"* (PMID: 42020136). Under-recognition in adults is a recurrent theme with important implications for surveillance and genetic counseling.

Finding 5 — COMT haploinsufficiency and Val158Met genotype modulate cognition and psychosis

COMT (catechol-O-methyltransferase), which degrades synaptic dopamine, lies within the deleted region. 22q11.2DS samples show *"an approximately 50% reduction in COMT mRNA, protein, and enzyme activity levels"* (PMID: 23992923). Because only one COMT allele remains, the functional **Val158Met (rs4680)** polymorphism on the intact allele has an outsized effect. In a cohort of 92 patients, *"22q11.2DS subjects with Met allele displayed higher IQ scores in all age* *groups compared to Val carriers, reaching significance in both adolescents and adults. The Met allele carriers performed better than Val carriers in EF tasks"* (PMID: 24853458). A longitudinal imaging study further found that COMT-Met carriers showed greater dorsal prefrontal cortex gray-matter reduction over time, linking COMT genotype to structural brain trajectories predicting psychosis (PMID: 20817203).

- **Ontology:** Gene HGNC:2228 (COMT); CHEBI:18243 (dopamine); GO:0042420 (dopamine catabolic process).

Finding 6 — Chromatin-regulatory modifier genes influence conotruncal heart defect risk

Conotruncal CHD occurs in **40–50% of patients**, but not all deletion carriers develop it — pointing to modifiers. Whole-genome sequencing of **456 conotruncal-defect cases vs 537 controls (all with 22q11.2DS)** identified rare damaging variants in **37 chromatin regulatory genes** (including **EP400, KAT6A, KMT2C, KMT2D, NSD1, CHD7, PHF21A, EZH2**): *"we identified 37 chromatin regulatory genes, that may increase risk for conotruncal heart defects in 8.5% of 22q11.2DS cases"* (PMID: 37463940). These genes are co-expressed with TBX1 in cardiac progenitor cells and genetically interact with Tbx1 in mice.

Direct experimental confirmation comes from a mouse model: *"conditional inactivation of Kmt2d in the Tbx1 lineage in Tbx1-heterozygous mice leads to fully penetrant perinatal lethality and increased incidence of craniofacial dysmorphism, thymus and parathyroid gland hypoplasia, and aortic arch anomalies"* (PMID: 42635044). This establishes **KMT2D as a bona fide genetic modifier** of TBX1 haploinsufficiency.

- **Ontology:** GO:0006338 (chromatin remodeling); Genes HGNC:7133 (KMT2D), HGNC:15851 (KMT2C).

Finding 7 — Congenital heart disease is the leading cause of pediatric mortality; mouse models recapitulate core features

"Congenital heart diseases (CHDs) and cardiovascular abnormalities are one of the pillars of clinical diagnosis of 22q11.2 deletion syndrome (22q11.2DS) and still represent the main cause of mortality in the affected children" (PMID: 29663641). As survival improves, the adult population has grown, bringing acquired cardiovascular problems and multisystem comorbidity into focus.

Tbx1-haploinsufficient mice recapitulate the syndrome's core features — craniofacial anomalies, thymus/parathyroid hypoplasia, and aortic arch/conotruncal anomalies — and also CNS phenotypes: *"In mice, this paraflocculus/flocculus dysplasia is associated with haploinsufficiency of the Tbx1 gene. Tbx1 haploinsufficiency also leads to impaired cerebellar synaptic plasticity and motor learning"* (PMID: 39638997).

Finding 8 — Autosomal dominant, mostly de novo (~90%), with ~50% offspring transmission risk

"The majority of deletions are de novo, with 10% or less inherited from an affected parent" (PMID: 14708107). The recurrent deletion arises via **non-allelic homologous recombination (NAHR)** between the low-copy-repeat blocks LCR22A–D. Inheritance is **autosomal dominant** — an affected individual has a **~50% chance of transmitting** the deletion. Recurrence risk to unaffected parents is low; independent de novo events have been documented even in first cousins, and germline mosaicism is rare.

Finding 9 — Diagnosis by CMA/FISH/MLPA; prenatal cfDNA with fetal-fraction amplification achieves high PPV

The small size of the microdeletion makes standard prenatal cell-free DNA (cfDNA) screening technically challenging. A cfDNA assay incorporating **fetal-fraction amplification** screened **379,428 patients**; of 76 screen-positives for de novo 22q11.2 microdeletion, *"22 (29.7%) had diagnostic testing results available, and all 22 cases were confirmed as true positives, for a PPV of 100% (95% CI 84.6%-100%)"* (PMID: 38622914). A multiplex digital-PCR cffDNA assay achieved 73.3% sensitivity / 96.5% specificity (PMID: 37684689). Current professional guidance (SMFM Consult Series #74, 2025) is cautious: *"we do not recommend routine general population screening for any microdeletion condition. Patients who choose to undergo cfDNA screening for 22q11.2 deletion specifically should do so only after appropriate pretest counseling"* (PMID: 42597105).

Finding 10 — Thymic hypoplasia causes T-cell immunodeficiency, autoimmunity, and atopy; complete athymia requires thymus transplant

Impaired thymus development is central to the immune phenotype: *"This thymic dysfunction impairs T-cell development and maturation, leading to a spectrum of T-cell lymphopenia, a restricted T-cell receptor repertoire and defects relating to regulatory T cells. Consequently, patients exhibit increased susceptibility to recurrent infections, particularly of the respiratory tract and a higher prevalence of autoimmune conditions and atopic diseases"* (PMID: 41730804).

Most patients have **partial DiGeorge** — mild-to-moderate T-cell lymphopenia that tends to improve with age. A small subset (**<1%**) have **complete athymia**, for which *"Thymus implantation is the definitive treatment"* (PMID: 41902846); allogeneic hematopoietic cell transplantation is an alternative. Impaired T-cell surveillance can permit EBV-associated complications, including chronic facial granulomatous dermatitis and, rarely, early Hodgkin lymphoma (PMID: 42114650; PMID: 41942950).

- **Ontology:** HP:0002721 (immunodeficiency); HP:0005359 (thymic hypoplasia); CL:0000084 (T cell); CL:0000815 (regulatory T cell); UBERON:0002370 (thymus).

Finding 11 — Parathyroid/thyroid dysgenesis causes hypocalcemia and thyroid dysfunction; hemizygosity can unmask recessive disease

"Individuals may also present with hypocalcemia and thyroid dysfunction due to impaired parathyroid gland formation and thyroid dysgenesis, respectively" (PMID: 40038168). Neonatal hypocalcemia may present with seizures or tetany and can be refractory to conventional therapy (PMID: 40866089). Uniquely, hemizygosity of the region can **unmask recessive disease** from a variant on the non-deleted homolog: *"Variations in genes on the non-deleted homologous chromosome in the critical 22q11.2 region can further influence phenotypes by revealing recessive diseases. This effect has been documented for several genes in this area, such as SNAP29 and GP1BB"* (PMID: 40038168) — e.g., CEDNIK syndrome (SNAP29) and Bernard-Soulier syndrome (GP1BB).

- **Ontology:** HP:0002917 (hypocalcemia); HP:0000829 (hypoparathyroidism); UBERON:0001132 (parathyroid gland); UBERON:0002046 (thyroid gland).

Finding 12 — 22q11.2DS markedly elevates the risk of early-onset Parkinson's disease

In a multicenter study of **856 adults** with 22q11.2DS: *"PD was present in 1.8% (95% CI: 0.9-2.6%) of the sample, 3.4% (95% CI: 2.2-4.6%) when including uncertain PD (clinical diagnosis or suspicion, but not meeting all criteria), and 14.0% (95% CI: 6.9-21.0%) of those aged ≥50 years. Median age at motor onset was 45 (range 20-66) years"* (PMID: 39918054). This early onset (median ~45 years vs ~60 in idiopathic PD) has prompted proposals for periodic motor evaluation from age 40. A candidate biomarker exists: **salivary monomeric α-synuclein** is reduced in parkinsonian 22q11.2DS patients (AUC 0.86 discriminating Park+ from Park−)

(PMID: 38661486). A Del(3.0 Mb)/+ mouse shows *"reduced axon innervation of dopaminergic neurons into the prefrontal cortex"* (PMID: 37238632), a dopaminergic deficit relevant to both parkinsonian and psychiatric phenotypes.

- **Ontology:** HP:0001300 (parkinsonism); CL:0000700 (dopaminergic neuron); CHEBI:18243 (dopamine).

Detailed Section Coverage

2. Etiology

- **Primary cause (genetic):** Hemizygous recurrent deletion at 22q11.2, most often the ~3 Mb LCR22A–D region (~106 genes); TBX1 haploinsufficiency is the dominant driver of physical phenotypes.
- **Genetic risk factors:** The deletion itself is causal and highly penetrant for *having the syndrome*, though individual features are variably expressed. **Modifier genes** shape severity: rare damaging variants in **chromatin regulators** (KMT2D, KMT2C, EP400, NSD1, CHD7, EZH2, etc.) increase conotruncal CHD risk in ~8.5% of cases (PMID: 37463940); **COMT Val158Met** and **PRODH** variants modify cognition/psychosis (PMID: 24853458); genes on the non-deleted homolog can act as protective (e.g., a ZFPM2 variant) or unmask recessive disease (PMID: 25981510; PMID: 40038168).
- **Environmental risk factors:** None established as necessary — the syndrome is fundamentally genetic. NAHR at the LCR22 repeats occurs sporadically; advanced parental age is not a robust risk factor for this recurrent CNV. Prenatal alcohol exposure and 22q11.2 CNVs may co-occur incidentally in neurodevelopmental cohorts (PMID: 42545840), but no causal environmental trigger is defined.
- **Protective factors:** A ZFPM2 variant downstream of TBX1 was proposed as protective against Tetralogy of Fallot in one proband (PMID: 25981510); COMT-Met alleles are relatively protective for cognition.
- **Gene–environment interactions:** Not well characterized; the phenotype is largely genetically determined with modifier-gene rather than environmental modulation.

3. Phenotypes

Phenotype	Type	Frequency	Onset	HPO term
Conotruncal congenital heart disease (TOF, truncus arteriosus, IAA, VSD)	Physical malformation	40–50%	Congenital	HP:0001636 / HP:0001671
Velopharyngeal insufficiency / cleft palate	Physical/ functional	~65–70% (palatal)	Congenital/ childhood	HP:0000220 / HP:0000175
T-cell immunodeficiency / recurrent infection	Laboratory/ clinical	~75% (mostly mild)	Neonatal/ infancy	HP:0005387 / HP:0002719
Hypocalcemia / hypoparathyroidism	Laboratory	~50–65% (often neonatal, may recur)	Neonatal	HP:0002917 / HP:0000829
Intellectual disability / learning disability	Behavioral/ cognitive	Majority; borderline-mild	Childhood	HP:0001249
Schizophrenia / psychosis	Behavioral	20–30% lifetime	Adolescence/ adulthood	HP:0100753
Autism spectrum / ADHD	Behavioral	Common	Childhood	HP:0000717 / HP:0007018
Parkinsonism / early-onset PD	Clinical sign	1.8% overall; 14% if ≥50 y	Adult (median 45 y)	HP:0001300
Characteristic craniofacial dysmorphism	Physical	Common	Congenital	HP:0001999
Chronic otitis media / hearing loss / airway anomalies	Clinical	High ENT burden	Childhood	HP:0000388 / HP:0000365

Expressivity is **highly variable** even within families; severity ranges from mild to life-threatening. Quality-of-life impact is substantial and lifelong, spanning cardiac, immune, endocrine, cognitive, psychiatric, speech, and ENT domains, requiring multidisciplinary care and structured pediatric-to-adult transition (PMID: 40223294; PMID: 41800660).

4. Genetic / Molecular Information

- **Causal locus/genes:** 22q11.2 deletion; **TBX1** (HGNC:11592; OMIM *602054) principal driver. Other dosage-sensitive genes: COMT (HGNC:2228), DGCR8 (microprocessor/miRNA biogenesis), CRKL, PRODH, SEPTIN5, RTN4R, ZDHHC8**.
- **Variant class:** Structural — recurrent **copy-number deletion** (LCR22A–D ~3 Mb; LCR22A–B ~1.5 Mb; nested/atypical deletions occur). Not a point mutation; functional consequence is **loss of function via haploinsufficiency**.
- **Allele frequency:** As a recurrent pathogenic CNV, birth prevalence ~1/4,000; depleted in healthy adult population databases (gnomAD-SV) consistent with strong selection.
- **Origin:** Predominantly **germline de novo** (~90%); ~10% inherited (autosomal dominant).
- **Modifier genes:** Chromatin regulators (KMT2D etc.), COMT/PRODH, and non-deleted-homolog variants (Findings 5, 6, 11).
- **Chromosomal abnormality:** Interstitial microdeletion of 22q11.21; visible by FISH/microarray, not by standard karyotype in most cases.
- **Related dosage disorder:** 22q11.21 dosage has been linked to Alzheimer risk in large exome analyses, narrowing to a SCARF2-KLHL22-MED15 region (PMID: 42648288) — illustrating the locus's broad neurologic relevance.

5. Environmental Information

No infectious agent or environmental toxin causes 22q11.2DS; it is a constitutional genetic disorder. Secondary/opportunistic infections (viral, especially **EBV**; recurrent respiratory pathogens) arise as *consequences* of T-cell immunodeficiency rather than causes of the syndrome (PMID: 41730804; PMID: 41942950). Lifestyle factors do not contribute to etiology.

6. Mechanism / Pathophysiology — Ordered Causal Chain

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1. Recurrent NAHR between LCR22A–D low-copy repeats
      → results in a de novo ~3 Mb hemizygous 22q11.2 deletion (~106 genes)      [PMID 32117416,
2. Deletion → haploinsufficiency of dosage-sensitive genes, chiefly TBX1        [PMID 23799583]
3. Reduced TBX1 dosage in pharyngeal ecto/endo/mesoderm
      → dysregulated FGF, retinoic-acid, BMP, and WNT/β-catenin signaling        [PMID 23799583]
4. Aberrant signaling to the rostral neural crest (non-cell-autonomous)
      → defective pharyngeal-arch and neural-crest development                    [PMID 2379958
      ├── BRANCH A → 4th-arch artery / outflow-tract defects
      │        → conotruncal CHD (TOF, truncus, IAA-B)                            [PMID 4003816
      │        → leading cause of pediatric mortality
      ├── BRANCH B → 3rd/4th pouch (thymus) hypoplasia
      │        → impaired T-cell development, restricted TCR repertoire, Treg defect
      │        → immunodeficiency, autoimmunity, atopy, EBV complications         [PMID 4173080
      ├── BRANCH C → parathyroid (3rd/4th pouch) + thyroid dysgenesis
      │        → hypoparathyroidism → hypocalcemia (seizures/tetany)              [PMID 4003816
      ├── BRANCH D → palatal / velopharyngeal + craniofacial maldevelopment
      │        → cleft palate, VPI, dysmorphism, ENT disease                      [PMID 4003816
      └── BRANCH E (neuro) → COMT/DGCR8/PRODH dosage + interneuron vulnerability
               → ~50% ↓ COMT activity → altered prefrontal dopamine              [PMID 23992923]
               → VIP+ CGE interneuron bias; reduced DA innervation of PFC
               → intellectual disability, 20–30% schizophrenia, early-onset PD    [PMID 4023613
5. Genetic modifiers (chromatin regulators e.g. KMT2D; COMT Val158Met;
   non-deleted-homolog variants) tune severity/penetrance of each branch         [PMID 37463940,
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Upstream vs downstream: The deletion and TBX1 haploinsufficiency are the most upstream events; pharyngeal/neural-crest signaling failure is the key developmental node from which the organ-specific branches diverge. The neuropsychiatric branch is partly independent, driven by dosage of COMT/DGCR8/PRODH and cortical interneuron vulnerability rather than by TBX1 alone — consistent with evidence that the physical phenotype is largely TBX1-attributable while the brain phenotype is **multigenic** (PMID: 34552467). Changes in brain metabolism from deleted metabolic genes may further contribute to variable brain phenotypes (PMID: 39567195).

- **Cell types (CL):** migratory cranial neural crest cell (CL:0000333), T cell (CL:0000084), regulatory T cell (CL:0000815), VIP GABAergic interneuron (CL:4023070), dopaminergic neuron (CL:0000700), cardiac progenitor cell.
- **Biological processes (GO):** neural crest cell development (GO:0014032); outflow tract morphogenesis (GO:0003151); thymus development (GO:0048538); parathyroid gland development (GO:0060017); dopamine catabolic process (GO:0042420); chromatin remodeling (GO:0006338).

7. Anatomical Structures Affected

- **Cardiovascular:** heart outflow tract, aortic arch / 4th arch arteries (UBERON:0004145), great vessels.
- **Immune/endocrine:** thymus (UBERON:0002370), parathyroid glands (UBERON:0001132), thyroid (UBERON:0002046).
- **Craniofacial/upper aerodigestive:** palate (UBERON:0001716), velopharynx, middle/inner ear, larynx/airway; characteristic facies.
- **Nervous system:** cerebral cortex (prefrontal/dorsal PFC), cerebellum (paraflocculus/flocculus), widespread reductions in brain volume and cortical surface area with increased cortical thickness (PMID: 41358290).
- **Subcellular/molecular compartments:** nucleus (transcription-factor and chromatin dysfunction), synapse (dopaminergic/cerebellar synaptic plasticity), mitochondria/metabolism.
- **Lateralization:** cardiac and aortic-arch anomalies are frequently right-sided/laterality defects; craniofacial features are typically bilateral.

8. Temporal Development

- **Onset:** Congenital (developmental malformations present at birth); immunodeficiency and hypocalcemia manifest neonatally; psychiatric features emerge in adolescence/adulthood; parkinsonism in mid-adulthood (median 45 y).
- **Course:** Chronic, lifelong. Partial-DiGeorge immune function tends to **improve with age**; hypocalcemia can be **episodic/recurrent** (latent hypoparathyroidism unmasked by stress). Psychiatric and neurodegenerative risks are **progressive** across the lifespan.
- **Critical periods:** Embryonic pharyngeal-arch development (weeks 4–8) for structural defects; adolescence for psychosis risk (surveillance window); age ≥40 for parkinsonism surveillance.

9. Inheritance and Population

- **Prevalence:** ~1 in 4,000 live births (~25/100,000); most common microdeletion syndrome. Likely under-ascertained in adults.
- **Inheritance:** Autosomal dominant; ~90% de novo, ~10% inherited; ~50% transmission risk from an affected parent.
- **Penetrance/expressivity:** Essentially complete penetrance for *having some phenotype* but **highly variable expressivity**; no single feature is obligate.

- **Anticipation:** Not a repeat-expansion disorder; no true anticipation, though ascertainment bias can mimic it.
- **Germline mosaicism:** Rare; recurrence risk to unaffected parents is low.
- **Founder effect / consanguinity:** Not applicable — recurrent de novo CNV mediated by ubiquitous LCR22 architecture, not population-specific founder alleles; occurs across all ethnic groups.
- **Sex ratio:** Approximately equal (autosomal).

10. Diagnostics

- **First-line molecular diagnosis:** Chromosomal microarray (CMA) — highest resolution for deletion size; **FISH** (TUPLE1/N25 probes) and **MLPA** are widely used and can detect the recurrent deletion. Standard karyotype is usually normal.
- **Prenatal:** cfDNA screening feasible; high PPV with fetal-fraction amplification (PPV 100% in one study, PMID: 38622914); multiplex dPCR cffDNA 73.3%/96.5% sens/spec (PMID: 37684689); confirmatory diagnostic testing (CVS/amniocentesis with CMA/FISH) required. Routine population microdeletion cfDNA screening is not recommended (PMID: 42597105).
- **Ancillary labs/imaging:** Serum calcium/PTH; immune panel (T-cell subsets, TREC on newborn screening for SCID/athymia detection); echocardiography; renal ultrasound; brain MRI for research/neurologic indications; panoramic radiograph trabecular analysis as a research bone-health biomarker (PMID: 42277781); salivary α-synuclein as a candidate parkinsonism biomarker (PMID: 38661486).
- **Differential diagnosis:** CHARGE syndrome, other conotruncal-anomaly/DiGeorge-spectrum causes (e.g., TBX1 point mutations, chromosome 10p13 deletion), Smith-Magenis syndrome, and other neurodevelopmental CNV syndromes.

11. Outcome / Prognosis

- **Mortality:** Congenital heart disease is the **leading cause of death in children** (PMID: 29663641); complete athymia is life-threatening without transplant.
- **Life expectancy:** Reduced but improving; a growing adult population faces acquired cardiovascular disease, psychiatric illness, and neurodegeneration (PMID: 40223294).
- **Morbidity/function:** Substantial — cognitive, psychiatric, speech, ENT, immune, and endocrine burdens impair daily functioning; negative psychiatric symptoms link to poorer global and social functioning (PMID: 42585902).

- **Prognostic factors:** Severity of CHD; presence of complete athymia; COMT genotype and modifier burden for neurocognition; smaller cerebellar volume relative to syndrome peers predicts lower language scores (PMID: 41358290).

12. Treatment

No cure exists; management is **organ-directed and multidisciplinary**, per 22q11.2 Society consensus guidelines for children (PMID: 36729053) and adults (PMID: 36729052).

Domain	Intervention	NCIT-type term
Cardiac	Surgical repair of conotruncal defects	Cardiac surgery
Endocrine	Calcium + active vitamin D (calcitriol); teriparatide for refractory hypocalcemia (PMID: 40866089)	Calcium/vitamin D supplementation
Immune	Infection prophylaxis, irradiated/CMV-safe blood, live-vaccine caution; **cultured thymus tissue implantation** for complete athymia; allogeneic HCT alternative (PMID: 41902846; PMID: 39303894)	Thymus transplantation
Psychiatric	Antipsychotics, standard psychiatric care; caution given COMT dosage	Antipsychotic therapy
ENT/ speech	Palatoplasty, VPI surgery, otologic/airway management, speech therapy (PMID: 41800660)	Speech therapy / palatoplasty
Supportive	Developmental, educational, and rehabilitative services	Rehabilitation therapy

Immunological management is standardized in dedicated guidelines (PMID: 36648576). Pharmacogenomic caution around COMT-metabolized drugs is biologically plausible but not yet guideline-mandated. No approved gene/RNA therapy exists; mouse studies have shown pharmacological rescue of single-gene cortical phenotypes as proof of concept (PMID: 34552467).

13. Prevention

- **Primary prevention:** Not possible (constitutional de novo genetic event). Genetic counseling for affected adults (50% transmission risk); reproductive options include preimplantation genetic testing (PGT) and prenatal diagnosis.

- **Secondary prevention:** Newborn SCID/TREC screening detects athymia; early echocardiography and calcium monitoring enable timely intervention.
- **Tertiary prevention:** Structured lifelong surveillance per consensus guidelines to prevent complications (cardiac, endocrine, immune, psychiatric); proposed motor evaluation from age 40 for parkinsonism.
- **Counseling:** Genetic counseling is central (recurrence risk, transmission, variable expressivity).

14. Other Species / Natural Disease

- **Taxonomy/orthologs:** *Tbx1* is conserved (mouse *Tbx1*, NCBI Gene 21380; human TBX1 NCBI Gene 6899). No naturally occurring 22q11.2DS-equivalent contiguous deletion is described in companion animals; the disorder is modeled experimentally rather than occurring naturally. Evolutionary conservation of the TBX1 pharyngeal program underpins cross-species model validity.

15. Model Organisms

- **Mouse (primary model):** *Tbx1* heterozygous and Df(16)A / Del(3.0Mb) deletion mice recapitulate craniofacial anomalies, thymus/parathyroid hypoplasia, aortic-arch/conotruncal anomalies, cerebellar paraflocculus/flocculus dysplasia with motor-learning deficits, and reduced prefrontal dopaminergic innervation (PMID: 39638997; PMID: 37238632). *Kmt2d;Tbx1* compound models demonstrate modifier gene action (PMID: 42635044).
- **Model types available:** knockout, conditional/lineage-specific, and deletion (Df) mice.
- **Strengths:** Faithfully reproduce physical malformations and specific neurobehavioral endophenotypes; enable dissection of dosage and modifier effects, and pharmacological rescue proof-of-concept (PMID: 34552467).
- **Limitations:** The multigenic human brain phenotype (schizophrenia, complex cognition) is only partially captured; species differences in gene content of the syntenic region and in cortical interneuron biology limit full recapitulation.

Mechanistic Model / Interpretation

The unifying interpretation is that **a single upstream lesion — the 22q11.2 deletion — produces a fan of downstream phenotypes through two partly separable routes.** The **TBX1-dependent route** explains the "physical" DiGeorge/VCFS features: reduced TBX1 dosage perturbs FGF/RA/BMP/WNT signaling in the pharyngeal apparatus and its cross-talk with neural crest, and the affected pharyngeal pouches/arches map cleanly onto the affected organs (outflow tract, thymus, parathyroid, palate, ear/airway). The **TBX1-independent, multigenic route** explains the "brain" phenotypes: dosage of COMT, DGCR8, PRODH and others, together with intrinsic vulnerability of VIP+ CGE-derived interneurons and reduced prefrontal dopaminergic innervation, drives intellectual disability, the 20–30% schizophrenia risk, and early-onset parkinsonism. Overlaid on both routes is a **modifier layer** — chromatin regulators (KMT2D and network), COMT Val158Met, and variants unmasked on the intact homolog — that converts uniform haploinsufficiency into the syndrome's hallmark **variable expressivity**. This two-route-plus-modifier architecture reconciles why physical features are largely TBX1-attributable and highly recurrent across models, while neuropsychiatric outcomes are heterogeneous and only partially modeled in mice.

Evidence Base

PMID	Contribution	Evidence type
23799583	TBX1 as principal driver; pharyngeal/neural-crest signaling	Human/model review
32117416	Deletion size (~3 Mb), ~106 genes	Review
35645294	Prevalence ~1/4000; deletion sizes; craniofacial genetics	Review
40038168	Neural crest → CHD/palate; hypocalcemia/thyroid; recessive unmasking	Review
42585902	20–30% schizophrenia risk; negative symptoms	Human cohort
40236134	VIP+ CGE interneuron vulnerability	Genomic/cellular
42020136	4.3% of TOF carry deletion; adult under-recognition	Human cohort
23992923	~50% ↓ COMT mRNA/protein/activity; psychosis haplotypes	Human/molecular
24853458	COMT Val158Met modifies IQ/EF (n=92)	Human cohort
37463940	37 chromatin regulators modify CHD (8.5%)	Human WGS
42635044	Kmt2d modifies Tbx1 haploinsufficiency	Mouse model
29663641	CHD as leading pediatric mortality	Review
39638997	Tbx1 cerebellar/motor phenotype	Mouse model
14708107	~90% de novo; independent de novo events	Human case
38622914	cfDNA fetal-fraction amplification PPV 100%	Human screening
42597105	SMFM cfDNA guidance	Guideline
41730804	Thymic dysfunction → immunodeficiency/autoimmunity/atopy	Review
41902846	Thymus implantation for athymia	Human cohort
39918054	PD prevalence; onset median 45 y (n=856)	Human cohort
37238632	Reduced PFC dopaminergic innervation	Mouse model

Limitations and Knowledge Gaps

1. **No primary data analysis was performed** — this report synthesizes literature and curated database identifiers; frequency figures are approximate and drawn from heterogeneous cohorts.
2. **Ontology IDs (MONDO, OMIM, HPO, GO, CL, UBERON, CHEBI, NCIT) were assigned from domain knowledge**, not programmatically verified against current ontology releases; they should be cross-checked before database ingestion.
3. **Frequency/penetrance estimates vary** by ascertainment (cardiac clinics vs psychiatric vs population), and adult under-recognition biases prevalence downward.
4. **Neuropsychiatric mechanism remains incompletely resolved** — the relative contributions of COMT, DGCR8, PRODH, interneuron biology, and metabolic changes are not fully quantified, and human mechanistic evidence is largely correlational.
5. **Biomarkers (salivary α-synuclein, mandibular trabecular FD) are exploratory**, based on small samples (n≈10–11 per group) and require validation.
6. **Treatment evidence base is limited** — consensus guidelines are largely expert-opinion given the rarity and heterogeneity of the syndrome; no disease-modifying therapy exists.

Proposed Follow-up Experiments / Actions

1. **Verify all ontology term IDs** against current MONDO/HPO/GO/CL/UBERON/NCIT releases before knowledge-base ingestion.
2. **Assemble a per-phenotype frequency table with confidence intervals** from the largest available registries (22q11.2 Society, ENIGMA-22q, International 22q11.2 Brain Behavior Consortium) rather than narrative-review estimates.
3. **Systematically catalog modifier variants** (chromatin regulators, COMT/PRODH, non-deleted-homolog variants) with effect sizes for genotype–phenotype annotation.
4. **Validate candidate biomarkers** — salivary α-synuclein for parkinsonism and imaging-derived cerebellar/cortical deviation scores for cognitive prognosis — in larger prospective cohorts.
5. **Curate the mouse-model phenotype-recapitulation matrix** (Tbx1, Df(16)A, Del(3.0Mb), Kmt2d compounds) mapping each mouse phenotype to its human HPO counterpart with recapitulation confidence.

6. **Track emerging trials** (ClinicalTrials.gov) for thymus tissue implantation, psychiatric interventions, and any gene/RNA-directed approaches, capturing NCT identifiers.

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